M2 + E17: Positivity and Singular-Limit Consistency of the Regularized Curvature Operator

Abstract

We prove that the corrected modulus-squared regularization of the WCT curvature-feedback operator is well-defined for all finite complex field values at every fixed regularization scale $\varepsilon > 0$. The regularized reciprocal is smooth and globally bounded, converges to the ordinary reciprocal at every nonzero field value, and admits quantitative uniform and L^2 convergence estimates away from field nodes. We also verify global $U(1)$ phase invariance and exhibit a counterexample to the historical, uncorrected denominator. These algebraic and analytic properties do not by themselves imply well-posedness or stability of a nonlinear evolution equation.

1 Definitions and theorem

Let $\Omega \subseteq \mathbb{R}^d$ be open, $\alpha \in \mathbb{R}$, and $\varepsilon > 0$. For $z \in \mathbb{C}$, define

$$D_\varepsilon(z) := |z|^2 + \varepsilon^2 e^{-2\alpha|z|^2}, \qquad R_\varepsilon(z) := \frac{\overline{z}}{D_\varepsilon(z)}. \tag{1}$$

For $\psi \in C^2(\Omega;\mathbb{C})$, define the regularized curvature operator

$$\Theta_\varepsilon[\psi](x) := -(\Delta\psi(x))R_\varepsilon(\psi(x)). \tag{2}$$

The Laplacian is applied componentwise to the complex-valued field.

Theorem 1 (M2 + E17)**.** *For every $\varepsilon > 0$ and $\alpha \in \mathbb{R}$ the following hold.*

1. *$D_\varepsilon(z) > 0$ for every $z \in \mathbb{C}$, including $z = 0$.*
2. *R_ε is a globally bounded C^∞ map $\mathbb{R}^2 \to \mathbb{R}^2$, identifying $\mathbb{C}$ with $\mathbb{R}^2$, and $R_\varepsilon(0) = 0$.*
3. *For every $z \neq 0$,*
$$\lim_{\varepsilon\to 0^+} R_\varepsilon(z) = \frac{1}{z}. \tag{3}$$
4. *If $\alpha \geq 0$, then for every $\delta > 0$,*
$$\sup_{|z|\geq\delta} \left| R_\varepsilon(z) - \frac{1}{z} \right| \leq \frac{\varepsilon^2}{\delta^3}. \tag{4}$$
For arbitrary $\alpha \in \mathbb{R}$, the convergence is uniform on every compact subset of $\mathbb{C} \setminus \{0\}$.
5. *For $\psi \in C^2(\Omega;\mathbb{C})$ and every $x \in \Omega$ such that $\psi(x) \neq 0$,*
$$\lim_{\varepsilon\to 0^+} \Theta_\varepsilon[\psi](x) = -\frac{\Delta\psi(x)}{\psi(x)}. \tag{5}$$

Proof. Step 1: Strict denominator positivity. For $z \in \mathbb{C}$, $|z|^2 \geq 0$. Because α and $|z|^2$ are real, $e^{-2\alpha|z|^2} > 0$. Multiplying by $\varepsilon^2 > 0$ gives

$$D_\varepsilon(z) = \underbrace{|z|^2}_{\geq 0} + \underbrace{\varepsilon^2 e^{-2\alpha|z|^2}}_{>0} > 0.$$

At the node $z = 0$, $D_\varepsilon(0) = \varepsilon^2 > 0$, and consequently $R_\varepsilon(0) = 0$.

Step 2: Smoothness and boundedness. Writing $z = x + iy$, the numerator $x - iy$ and the denominator

$$x^2 + y^2 + \varepsilon^2 e^{-2\alpha(x^2+y^2)}$$

are C^∞ in the real variables (x, y). Step 1 shows that the denominator is nowhere zero, so $R_\varepsilon \in C^\infty(\mathbb{R}^2; \mathbb{R}^2)$.

To prove a global bound, fix any $\rho > 0$ and set $\alpha_+ := \max\{\alpha, 0\}$. When $|z| \leq \rho$,

$$D_\varepsilon(z) \geq \varepsilon^2 e^{-2\alpha_+\rho^2}, \qquad |R_\varepsilon(z)| \leq \frac{\rho e^{2\alpha_+\rho^2}}{\varepsilon^2}.$$

When $|z| \geq \rho$, $D_\varepsilon(z) \geq |z|^2$, hence

$$|R_\varepsilon(z)| \leq \frac{|z|}{|z|^2} = \frac{1}{|z|} \leq \frac{1}{\rho}.$$

Therefore

$$\|R_\varepsilon\|_{L^\infty(\mathbb{C})} \leq \max\left\{\frac{1}{\rho}, \frac{\rho e^{2\alpha_+\rho^2}}{\varepsilon^2}\right\} < \infty. \tag{6}$$

This bound is for each fixed $\varepsilon > 0$; it is not asserted to be uniform as $\varepsilon \to 0^+$.

Step 3: Reciprocal limit. For fixed $z \neq 0$, the quantity $e^{-2\alpha|z|^2}$ is finite and independent of ε. Thus

$$\lim_{\varepsilon \to 0^+} D_\varepsilon(z) = |z|^2 > 0.$$

Continuity of division at a nonzero denominator yields

$$\lim_{\varepsilon \to 0^+} R_\varepsilon(z) = \frac{\overline{z}}{|z|^2} = \frac{\overline{z}}{z\overline{z}} = \frac{1}{z}.$$

Step 4: Quantitative convergence away from nodes. For $z \neq 0$, direct subtraction gives the exact identity

$$\begin{aligned} R_\varepsilon(z) - \frac{1}{z} &= \frac{1}{z}\left(\frac{|z|^2}{|z|^2 + \varepsilon^2 e^{-2\alpha|z|^2}} - 1\right) \\ &= -\frac{1}{z}\,\frac{\varepsilon^2 e^{-2\alpha|z|^2}}{|z|^2 + \varepsilon^2 e^{-2\alpha|z|^2}}. \end{aligned} \tag{7}$$

Therefore

$$\left|R_\varepsilon(z) - \frac{1}{z}\right| = \frac{\varepsilon^2 e^{-2\alpha|z|^2}}{|z|\left(|z|^2 + \varepsilon^2 e^{-2\alpha|z|^2}\right)} \leq \frac{\varepsilon^2 e^{-2\alpha|z|^2}}{|z|^3}. \tag{8}$$

If $\alpha \geq 0$ and $|z| \geq \delta > 0$, then the exponential is at most one. This proves (4).

For general α, let $K \subset \mathbb{C} \setminus \{0\}$ be compact. There exist $0 < \delta \le M < \infty$ such that $\delta \le |z| \le M$ on K. With $\alpha_- := \max\{-\alpha, 0\}$, we have

$$\sup_{z \in K} \left| R_\varepsilon(z) - \frac{1}{z} \right| \le \frac{\varepsilon^2 e^{2\alpha_- M^2}}{\delta^3} \xrightarrow[\varepsilon \to 0^+]{} 0.$$

Step 5: Curvature limit. Fix $x \in \Omega$ with $\psi(x) \neq 0$. Since ψ and $\Delta\psi$ are fixed independently of ε, Step 3 gives

$$\lim_{\varepsilon \to 0^+} \Theta_\varepsilon[\psi](x) = -(\Delta\psi(x)) \lim_{\varepsilon \to 0^+} R_\varepsilon(\psi(x)) = -\frac{\Delta\psi(x)}{\psi(x)}.$$

All the assertions follow. □

2 Convergence in L^2

Corollary 1 (Strong L^2 limit away from nodes)**.** *Assume $\alpha \ge 0$, $\psi \in H^2(\Omega; \mathbb{C})$, and $|\psi(x)| \ge \delta > 0$ for almost every $x \in \Omega$. Then*

$$\left\| \Theta_\varepsilon[\psi] + \frac{\Delta\psi}{\psi} \right\|_{L^2(\Omega)} \le \frac{\varepsilon^2}{\delta^3} \|\Delta\psi\|_{L^2(\Omega)} \xrightarrow[\varepsilon \to 0^+]{} 0. \tag{9}$$

Proof. By definition,

$$\Theta_\varepsilon[\psi] + \frac{\Delta\psi}{\psi} = (\Delta\psi) \left(\frac{1}{\psi} - R_\varepsilon(\psi) \right) \quad \text{a.e. on } \Omega.$$

Apply the pointwise multiplier estimate (4) with $z = \psi(x)$, then take the L^2 norm. □

Corollary 2 (Fixed-regularization L^2 control)**.** *If $\psi \in H^2(\Omega; \mathbb{C})$ and $\varepsilon > 0$ is fixed, then*

$$\|\Theta_\varepsilon[\psi]\|_{L^2(\Omega)} \le \|R_\varepsilon\|_{L^\infty(\mathbb{C})} \|\Delta\psi\|_{L^2(\Omega)} < \infty.$$

Proof. The multiplier $R_\varepsilon(\psi)$ is essentially bounded by (6), while $\Delta\psi \in L^2(\Omega)$. □

Remark 1 (Whole-space limitation)**.** *The hypothesis $|\psi| \ge \delta > 0$ almost everywhere on $\Omega = \mathbb{R}^d$ is incompatible with $\psi \in L^2(\mathbb{R}^d)$, since it would imply $\|\psi\|_2^2 \ge \delta^2 |\mathbb{R}^d| = \infty$. Thus Corollary 1 is not a blanket global convergence theorem for positive-mass localized whole-space fields.*

Remark 2 (No uniform convergence through a node)**.** *At $z = 0$, $R_\varepsilon(0) = 0$ whereas $1/z$ is undefined. The obstruction is quantitative: set $z = \varepsilon$ (real and positive). From* (8)*,*

$$\left| R_\varepsilon(\varepsilon) - \frac{1}{\varepsilon} \right| = \frac{e^{-2\alpha\varepsilon^2}}{\varepsilon(1 + e^{-2\alpha\varepsilon^2})} \sim \frac{1}{2\varepsilon} \quad (\varepsilon \to 0^+).$$

Therefore the reciprocal convergence is not uniform in any punctured neighborhood of the origin.

3 Global phase invariance

Proposition 1 (Constant $U(1)$ phase invariance)**.** *For every constant $\beta \in \mathbb{R}$ and $\psi \in C^2(\Omega;\mathbb{C})$,*

$$\Theta_\varepsilon[e^{i\beta}\psi] = \Theta_\varepsilon[\psi].$$

Proof. The denominator depends only on the modulus, so

$$R_\varepsilon(e^{i\beta}z) = e^{-i\beta}R_\varepsilon(z).$$

Since β is constant, $\Delta(e^{i\beta}\psi) = e^{i\beta}\Delta\psi$. Hence

$$\Theta_\varepsilon[e^{i\beta}\psi] = -e^{i\beta}(\Delta\psi)e^{-i\beta}R_\varepsilon(\psi) = \Theta_\varepsilon[\psi].$$

□

Remark 3. *Global $U(1)$ invariance does not by itself establish local gauge invariance under a position-dependent phase $\beta(x)$.*

4 Counterexample to the historical denominator

The earlier expression

$$\Theta_\varepsilon^{\text{old}}[\psi] = -\frac{\Delta\psi}{\psi + \varepsilon e^{-\alpha|\psi|^2}} \tag{10}$$

does not have a universally nonzero denominator. Indeed, take $\alpha = 0$ and $\psi \equiv -\varepsilon$. Then

$$\psi + \varepsilon e^{-\alpha|\psi|^2} = -\varepsilon + \varepsilon = 0.$$

The expression (10) is undefined there, even if $\Delta\psi = 0$ (which gives the indeterminate quotient $0/0$). By contrast, the corrected denominator satisfies

$$|\psi|^2 + \varepsilon^2 e^{-2\alpha|\psi|^2} > 0 \quad \text{at every finite complex field value.}$$

The historical denominator also fails generic constant-phase covariance because its two summands transform differently.

5 Scope of the result

The theorem establishes denominator positivity, pointwise regularity for fixed $\varepsilon > 0$, reciprocal consistency, quantitative convergence away from nodes, and global phase invariance. It does *not* by itself prove nonlinear PDE well-posedness, global-in-time existence, uniqueness, confinement, Lyapunov descent, or dynamical stability. Those require separate hypotheses and proofs for the chosen evolution.

References

[1] R. J. Reyes, *WCT Full Equation List (Corrected)*, `https://github.com/rickyjreyes/geometry_of_resonance/blob/main/WCT_FULL_EQUATION_LIST_CORRECTED.md`.

[2] R. J. Reyes, *WCT Master Equations (Updated)*, `https://github.com/rickyjreyes/geometry_of_resonance/blob/main/WCT_MASTER_EQUATIONS_UPDATED.md`.

[3] R. J. Reyes, *Derivation Closure*, `https://github.com/rickyjreyes/geometry_of_resonance/blob/main/DERIVATION_CLOSURE.md`.